Examiner only

12. (a) State what is meant by the enthalpy change of combustion. [2]

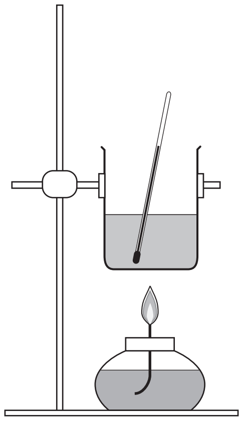
....

....

(b) (i) The equation that corresponds to the enthalpy change of combustion of propan-1-ol is shown.

$$C_3H_7OH(l) \quad + \quad 4\tfrac{1}{2}O_2(g) \longrightarrow 3CO_2(g) \quad + \quad 4H_2O(l)$$

The apparatus below can be used to determine the enthalpy change for this reaction.



A student used this apparatus and obtained the following results.

Initial mass of spirit burner with propan-1-ol = 126.16 g

Final mass of spirit burner with propan-1-ol = 126.02 g

Initial temperature of water = 21.0 °C

Final temperature of water = 33.5 °C

Volume of water in beaker = 100 cm^3

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Use these results in parts I and II.

I. Calculate the number of moles of propan-1-ol used up in the reaction. [1]

Number of moles = ... mol

II. Calculate the enthalpy change of combustion of propan-1-ol, $\Delta_c H$, under these conditions of temperature and pressure. [3]

$\Delta_c H$ = ... $kJ\,mol^{-1}$
sign *value*

(ii) Another student suggested that leaving the spirit burner to burn for longer would increase the accuracy of the experiment.

State whether you agree with this student. Explain your answer. [1]

....

....

(c) Enthalpy changes of combustion can be used to determine average bond enthalpies.

Some average bond enthalpies are given in the table.

Bond	Average bond enthalpy / kJ mol^{-1}
O=O	496
C—H	412
C—C	348
C=O	805
O—H	463

The equation that corresponds to the enthalpy change of combustion of butan-1-ol is shown below.

$$C_4H_9OH(l) \quad + \quad 6O_2(g) \longrightarrow 4CO_2(g) \quad + \quad 5H_2O(l) \qquad \Delta_c H = -2\,676\ \text{kJ mol}^{-1}$$

Examiner only

Use the enthalpy change of combustion of butan-1-ol and the average bond enthalpies in the table to calculate the average bond enthalpy of a C—O bond. [4]

Average bond enthalpy [C—O] = ... $kJ\,mol^{-1}$

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